

Лабораторная работа №1

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Тема: Система контроля версий Git.

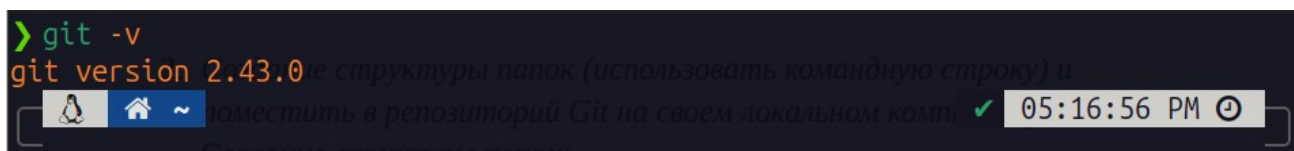
Цель: Познакомиться с возможностями и получить практические навыки работы с системой контроля версий Git.

Задание:

1) Подготовка рабочего окружения

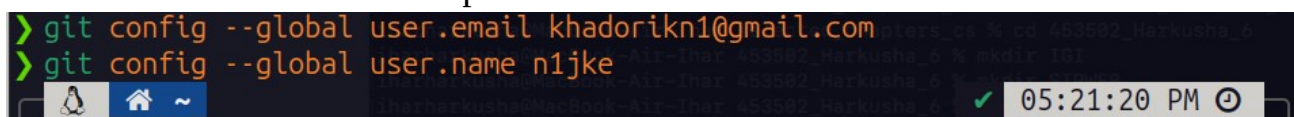
а. Установка Git

```
> git -v
git version 2.43.0
```



б. Выполнение базовых настроек

```
> git config --global user.email khadorikn1@gmail.com
> git config --global user.name n1jke
```



2) Создание структуры папок (использовать командную строку) и поместить в репозиторий Git на своем локальном компьютере

```
> mkdir 453503_Khadorik_27 && cd 453503_Khadorik_27
> ls
> mkdir IGI && cd IGI
> mkdir LR1
> mkdir LR2
> mkdir LR3
> mkdir LR4
> mkdir LR5
> cd .. && mkdir STRWEB && cd STRWEB
> mkdir LR1
> mkdir LR2
> mkdir LR3
> mkdir LR4
```



```

> cd LR1
> mkdir bar && cd bar && mkdir 705
> ls -foo/771
705 |--pwd/777
> cd .. |--test.txt/000
> mkdir foo && cd foo
> mkdir 233
> echo "007" > test.txt
> ls
233 test.txt
> cd .. --baz/744
> cd bar |--readme.txt/444
> mkdir bin && cd bin && mkdir 700
> ls -foo/111
700 --test.txt/000
> mkdir echo && cd echo && mkdir 555
> ls -bar/755
555 |--bin/700
> cd .. |--paste/555
> echo "400" > readme.txt
> cd .. --readme.txt/400
> mkdir baz && cd baz && mkdir 644
> mkdir example && cd example && mkdir 222

```

Результат:

```

> tree
.
├── bar
│   ├── 705
│   ├── baz
│   │   ├── 644
│   │   ├── example
│   │   └── 222
│   └── bin
│       ├── 700
│       ├── echo
│       └── 555
└── foo
    ├── 233
    └── test.txt

```

```

> git status
fatal: not a git repository (or any of the parent directories): .git
> git init
Initialized empty Git repository in /home/n1jke/453503_Khadorik_27/.git/
> git status
On branch main

No commits yet

Untracked files:
  (use "git add <file>..." to include in what will be committed)
    IGI/

nothing added to commit but untracked files present (use "git add" to track)

```

3) Создайте репозиторий внутри папки LR1 с помощью клонирования проекта с сайта [github.com](https://github.com/smartigaorg/geometric_lib) (https://github.com/smartigaorg/geometric_lib)

```
> git clone https://github.com/smartigaorg/geometric_lib
Cloning into 'geometric_lib'...
remote: Enumerating objects: 35, done.
remote: Total 35 (delta 0), reused 0 (delta 0), pack-reused 35 (from 1)
Receiving objects: 100% (35/35), 4.69 KiB | 4.69 MiB/s, done.
Resolving deltas: 100% (5/5), done.
```

4)

```
> git status
On branch main
Your branch is up to date with 'origin/main'.

nothing to commit, working tree clean
> git add circle.py square.py
> git status
On branch main
Your branch is up to date with 'origin/main'.

nothing to commit, working tree clean
```

a)

```
> echo "# some change" >> square.py
> cat square.py

def area(a):
    return a * a

def perimeter(a):
    return 4 * a
# some change
> git status
On branch main
Your branch is up to date with 'origin/main'.

Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
    modified:   square.py

no changes added to commit (use "git add" and/or "git commit -a")
```

b)

```
> git add square.py
> git commit -m "add changes"
[main 6e47851] add changes
1 file changed, 1 insertion(+)
> git status
On branch main
Your branch is ahead of 'origin/main' by 1 commit.
  (use "git push" to publish your local commits)

nothing to commit, working tree clean
```

c)

```
> touch test.go test.txt
> git status
On branch main
Your branch is ahead of 'origin/main' by 1 commit.
  (use "git push" to publish your local commits)

Untracked files:
  (use "git add <file>..." to include in what will be committed)
    test.go
    test.txt

nothing added to commit but untracked files present (use "git add" to track)
> rm -rf test.go test.txt
> git status
On branch main
Your branch is ahead of 'origin/main' by 1 commit.
  (use "git push" to publish your local commits)

nothing to commit, working tree clean
```

d)


```

> mv circle.py new_cirlce.py
> mv square.py square2.py
> git status
On branch main
Your branch is ahead of 'origin/main' by 1 commit.
  (use "git push" to publish your local commits)

Changes not staged for commit:
  (use "git add/rm <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
        deleted:    circle.py
        deleted:    square.py

Untracked files:
  (use "git add <file>..." to include in what will be committed)
        new_cirlce.py
        square2.py
no changes added to commit (use "git add" and/or "git commit -a")

```

e)

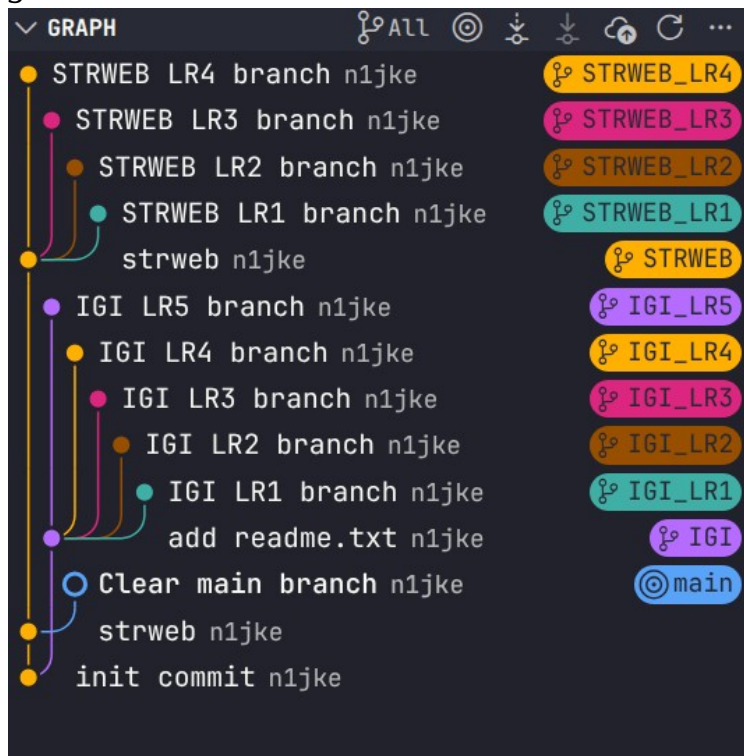
f)

`git switch STRWEB`

`git switch -c STRWEB_LR4`

`git rm -r STRWEB/LR1 STRWEB/LR2 STRWEB/LR3`

`git commit -m "STRWEB LR4 branch"`



g) `git log` - на текущей ветке

```

commit 760fabeb5b38f6dfde4ca3131afcee01992c8d9c (HEAD -> main)
Author: n1jke <khadorikn1@gmail.com>
Date: Mon Feb 16 19:34:07 2026 +0300

    Clear main branch

commit 7fe3dfdccc54ef3b70bee5dd93539ca9d7bd8b3d
Author: n1jke <khadorikn1@gmail.com>
Date: Mon Feb 16 19:24:48 2026 +0300

    strweb

commit 2ebd60682630eea03e7c4c126b759b24b333aefa
Author: n1jke <khadorikn1@gmail.com>
Date: Mon Feb 16 19:20:17 2026 +0300

    init commit
(END)

```

`git log --all` со всех веток

```
4d1bd71 (STRWEB_LR4) STRWEB LR4 branch
7932927 (STRWEB_LR3) STRWEB LR3 branch
e5d66fe (STRWEB_LR2) STRWEB LR2 branch
6c0d6ba (STRWEB_LR1) STRWEB LR1 branch
bce8164 (IGI_LR5) IGI LR5 branch
05f1483 (IGI_LR4) IGI LR4 branch
d716551 (IGI_LR3) IGI LR3 branch
0906775 (IGI_LR2) IGI LR2 branch
91956da (IGI_LR1) IGI LR1 branch
0a3dc88 (IGI) add readme.txt
```

`git log --all --oneline 3` - последние 3 коммита со всех веток в краткой форме

```
4d1bd71 (STRWEB_LR4) STRWEB LR4 branch
7932927 (STRWEB_LR3) STRWEB LR3 branch
e5d66fe (STRWEB_LR2) STRWEB LR2 branch
```

`git log --author="n1jke" --oneline` - коммиты по автору

```
760fabe (HEAD -> main) Clear main branch
7fe3dfd strweb
2ebd606 init commit
```

h) `git log -p - diff of commits`

```
commit 4d1bd719ed59f46bcf946a03788fb309b824336b (STRWEB_LR4)
Author: n1jke <khadorikn1@gmail.com>
Date: Mon Feb 16 19:43:48 2026 +0300

    STRWEB LR4 branch

diff --git a/STRWEB/LR1/readme.md b/STRWEB/LR1/readme.md
deleted file mode 100644
index 9daeafb..0000000
--- a/STRWEB/LR1/readme.md
+++ /dev/null
@@ -1,0,0 @@
-test
diff --git a/STRWEB/LR2/readme.md b/STRWEB/LR2/readme.md
deleted file mode 100644
index 9daeafb..0000000
--- a/STRWEB/LR2/readme.md
```

`--stat/shortstat` - statistic

```
commit 4d1bd719ed59f46bcf946a03788fb309b82433
Author: n1jke <khadorikn1@gmail.com>
Date: Mon Feb 16 19:43:48 2026 +0300

    STRWEB LR4 branch

STRWEB/LR1/readme.md | 1 -
STRWEB/LR2/readme.md | 1 -
STRWEB/LR3/readme.md | 1 -
3 files changed, 3 deletions(-)
```

`--named-only` - checked file names

`--name-status` - name + status(D, M, A)

`git log --all --pretty=...` - formatter

`--graph` - commits graph

i) *git show hash - diff between parent*

```
Commit 793292/acdc0fd23d18c73c0ecd169c3048572cd (STRWEB_LR3)
Author: n1jke <khadorikn1@gmail.com>
Date: Mon Feb 16 19:43:37 2026 +0300

STRWEB_LR3 branch

diff --git a/STRWEB/LR1/readme.md b/STRWEB/LR1/readme.md
deleted file mode 100644
index 9daeafb..0000000
--- a/STRWEB/LR1/readme.md
+++ /dev/null
@@ -1,0,0 @@
```

git diff hash^ hash - diff between commits

log -p - diff for all commits

```
> echo "27"> text.txt
> ls
text.txt
> git add test.txt
fatal: pathspec 'test.txt' did not match any
> git add text.txt
> git commit -m "add test.txt"
[main 78601e9] add test.txt
1 file changed, 1 insertion(+)
create mode 100644 text.txt
> git log --oneline
> git reset --hard HEAD~1
HEAD is now at 760fab6 Clear main branch
> git log --oneline
> echo "27" > text.txt
> git add text.txt
> git commit -m "add test.txt"
[main 41b39f9] add test.txt
1 file changed, 1 insertion(+)
create mode 100644 text.txt
> git reset --soft HEAD~1
> ls
text.txt
> git log --oneline
```

j)

--soft удаляет коммит, но сохраняет изменения

--hard - полностью удаляет коммит и изменения

git revert - еще можно, создаст новый коммит-отмену

```
> git add text.txt
> git status
On branch main
Changes to be committed:
  (use "git restore --staged <file>..." to unstage)
    new file:   text.txt
> git restore --staged text.txt
> git status
On branch main
Untracked files:
  (use "git add <file>..." to include in what will be committed)
    text.txt
```

k)


```

> git status
On branch main
nothing to commit, working tree clean
> echo "new" >> readme.md
> cat readme.md
27
new
> git status
On branch main
Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
        modified:   readme.md
no changes added to commit (use "git add" and/or "git commit -m")
> git restore readme.md
> cat readme.md
27
> git status
On branch main
nothing to commit, working tree clean

```

l)

5) Выполните следующие действия с удаленным репозиторием:

```

> git remote add origin https://github.com/n1jke/453503_Khadorik_27.git
git branch -M main
git push -u origin main
Enumerating objects: 19, done.
Counting objects: 100% (19/19), done.
Delta compression using up to 16 threads
Compressing objects: 100% (7/7), done.
Writing objects: 100% (19/19), 1.12 KiB | 575.00 KiB/s, done.
Total 19 (delta 0), reused 0 (delta 0), pack-reused 0
To https://github.com/n1jke/453503_Khadorik_27.git
 * [new branch]      main -> main
branch 'main' set up to track 'origin/main'.

```

a)

```

> git remote show origin
* remote origin
Fetch URL: https://github.com/n1jke/453503_Khadorik_27.git
Push URL: https://github.com/n1jke/453503_Khadorik_27.git
HEAD branch: main
Remote branches:
  IGI          tracked
  IGI_LR1      tracked
  IGI_LR2      tracked
  IGI_LR3      tracked
  IGI_LR4      tracked
  IGI_LR5      tracked
  STRWEB       tracked
  STRWEB_LR1   tracked
  STRWEB_LR2   tracked
  STRWEB_LR3   tracked
  STRWEB_LR4   tracked
  main         tracked
Local branches configured for 'git pull':
  IGI          merges with remote IGI
  IGI_LR1      merges with remote IGI_LR1
  IGI_LR2      merges with remote IGI_LR2
  IGI_LR3      merges with remote IGI_LR3
  IGI_LR4      merges with remote IGI_LR4
  IGI_LR5      merges with remote IGI_LR5
  STRWEB       merges with remote STRWEB
  STRWEB_LR1   merges with remote STRWEB_LR1
  STRWEB_LR2   merges with remote STRWEB_LR2
  STRWEB_LR3   merges with remote STRWEB_LR3
  STRWEB_LR4   merges with remote STRWEB_LR4
  main         merges with remote main
Local refs configured for 'git push':
  IGI          pushes to IGI (up to date)

```

b)

```

> git status
On branch main
Your branch is up to date with 'origin/main'.

Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
        modified:   readme.md

no changes added to commit (use "git add" and/or "git commit -a")
> git add readme.md
> git commit -m "upd readme.md"
[main bed8439] upd readme.md
 1 file changed, 1 insertion(+)
> git push origin main
Enumerating objects: 5, done.
Counting objects: 100% (5/5), done.
Writing objects: 100% (3/3), 249 bytes | 249.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0
To https://github.com/n1jke/453503_Khadorik_27.git
   e98d594..bed8439  main -> main

```

c)

```

> git remote -v
origin https://github.com/n1jke/453503_Khadorik_27.git (fetch)
origin https://github.com/n1jke/453503_Khadorik_27.git (push)
> git remote set-url origin https://github.com/n1jke/453503_Khadorik_
> git remote -v
origin https://github.com/n1jke/453503_Khadorik_27_labs.git (fetch)
origin https://github.com/n1jke/453503_Khadorik_27_labs.git (push)

```

d)

```

> git pull origin main
remote: Enumerating objects: 4, done.
remote: Counting objects: 100% (4/4), done.
remote: Compressing objects: 100% (3/3), done.
remote: Total 3 (delta 0), reused 0 (delta 0), pack
Unpacking objects: 100% (3/3), 217.98 KiB | 2.20 MiB/s, done.
From https://github.com/n1jke/453503_Khadorik_27_la
 * branch                main                -> FETCH_HEAD
    bed8439..1fb8483  main                -> origin/main
Updating bed8439..1fb8483
Fast-forward
 gopher.jpg | Bin 0 -> 222515 bytes
 1 file changed, 0 insertions(+), 0 deletions(-)
 create mode 100644 gopher.jpg

```

e)

or git fetch origin main

f) *git log --oneline --all -n 5*

```

1fb8483 (HEAD -> main, origin/main) add rnd picture
bed8439 upd readme.md
e98d594 add readme
4d1bd71 (origin/STRWEB_LR4, STRWEB_LR4) STRWEB LR4 branch
7932927 (origin/STRWEB_LR3, STRWEB_LR3) STRWEB LR3 branch
(END)

```



```

> git status
On branch 1
nothing to commit, working tree clean
> tree
.
├── bar
│   └── bin
│       └── readme.txt
└── foo
    ├── test.txt
    └── readme.txt
    └── test.txt

```

4 directories, 4 files

```

> git add bar
> git add foo
> git rm -rf foo/test.txt
rm 'IGI/LR1/foo/test.txt'
> git status
On branch 1
Changes to be committed:
  (use "git restore --staged <file>..." to unstage)
    deleted:    foo/test.txt

> git commit -m "5g) Delete file"
[172603e1] 5g) Delete file
1 file changed, 0 insertions(+), 0 deletions(-)
delete mode 100644 IGI/LR1/foo/test.txt

```

g)

```

> git checkout -b 2
Switched to a new branch '2'
> ls
google.png  gopher.jpg  main.go  readme.md
> git status
On branch 2
Untracked files:
  (use "git add <file>..." to include in what will be committed)
    google.png
    main.go

nothing added to commit but untracked files present (use "git add" to track)
> git add google.png main.go
> git commit -m "add new files"
[2 c9f5bf7] add new files
2 files changed, 7 insertions(+)
create mode 100644 google.png
create mode 100644 main.go
> git status
On branch 2
nothing to commit, working tree clean
> git switch main
Switched to branch 'main'
Your branch is up to date with 'origin/main'.
> git merge 2
Updating 1fb8483..c9f5bf7
Fast-forward
 google.png | Bin 0 -> 78670 bytes
main.go     | 7 +++++++
2 files changed, 7 insertions(+)
create mode 100644 google.png
create mode 100644 main.go
> git branch -D 2
Deleted branch 2 (was c9f5bf7).
> git branch --list

```

h)

6) https://n1jke.github.io/lr_page/